

IP-SAKTI SAHAYAK

Family 5 — Solanaceae / Ashwagandha

Objective: Create a TKDL-only family cluster around Ashwagandha for the proposed alternative-plant discovery feature. The system should retrieve related TKDL-listed plants, compare evidence, and rank them as potential alternatives for further research.

Strict TKDL rule: Plant candidates below are restricted to entries visible in the official TKDL plant-name index. No outside botanical database is used to add candidate names. Similarity scores are IP-SAKTI prototype ranking values—not scientific equivalence and not legal clearance.

1. Anchor / main plant

Ashwagandha — *Withania somnifera*

TKDL records the Ayurvedic name *ashwagandha*, Unani name *Asgand*, and several regional names. The public TKDL material also identifies *Withania coagulans* as another *Withania* species.

Key constituent terminology appearing in a TKDL-linked examination record: withanolide glycosides, withanolide aglycones and related naturally occurring constituents. This is recorded here only because the official TKDL-linked examination material discusses them; it should not be interpreted as a complete phytochemical profile.

2. Related TKDL species — 5 candidates

TKDL plant	Relationship	Similarity*	Evidence / role
<i>Withania coagulans</i> Dun.	Same genus; Solanaceae	VERY HIGH — 0.93	Direct TKDL-linked examination evidence for another <i>Withania</i> species; strong candidate
<i>Withania ashwagandha</i>	Same genus; Solanaceae	VERY HIGH — 0.90	Official TKDL plant index entry; identity/variant relationship needs verification
<i>Solanum indicum</i> Linn.	Same family; different genus	MEDIUM-HIGH — 0.72	Official TKDL plant index; traditional/formulation evidence should be matched
<i>Solanum nigrum</i> Linn.	Same family; different genus	MEDIUM-HIGH — 0.70	Official TKDL index + direct examination evidence
<i>Solanum surattense</i> Burm. F.	Same family; different genus	MEDIUM-HIGH — 0.70	Official TKDL index + direct formulation/examination evidence

*Prototype ranking only. A same-genus candidate starts higher; actual formulation/use overlap must raise or lower the final score.

3. TKDL plant-index verification

Official TKDL plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official TKDL index contains *Withania ashwagandha*, *Withania coagulans* and *Withania somnifera*. It also contains the *Solanum* entries used here, including *Solanum indicum*, *Solanum nigrum* and *Solanum surattense*.

4. TKDL evidence for Ashwagandha / *Withania*

EP1906980 — EPO: TKDL third-party observations brought prior-art references on *Withania somnifera* (*Ashwagandha*) for hyperacidity, gastric ulcer, anaemia, insomnia, convulsion, diabetes mellitus, excessive perspiration, restlessness and depression. The report identifies several TKDL database formulations, including *Safoof-e-Asgandh* 1, *Amukkara Karpam* and *Phalasava*. It records TKDL evidence of *Withania somnifera* root-based formulations and reports that specified claims were not considered novel; the application was later deemed withdrawn.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinerreport/reports/EP1906980.asp>

3507/MUM/2013 — Patent Office: The examination material states that *Withania somnifera* is widely used in Ayurveda and describes *Withania coagulans* as another species of *Withania* with reported hypoglycemic activity. The examiner concluded that the claimed subject matter lacked inventive step in light of the cited teachings.

Official TKDL examination PDF: https://www.tkdil.res.in/tkdil/Outcomes/3507-MUM-2013_Examination.pdf

5691/CHE/2015 — Patent Office: The examination material discusses *Ashwagandha* root extracts, active constituents and traditional uses, including tonic/rejuvenating and other uses, in the context of patent examination.

Official TKDL examination PDF: https://www.tkdil.res.in/tkdil/Outcomes/examination_report_5691CHE2015.pdf

5. Related-species evidence

***Withania coagulans*:** This is the strongest related candidate because the official TKDL-linked 3507/MUM/2013 examination material explicitly describes it as another species of *Withania* and discusses its reported activity alongside *Withania somnifera*. This is stronger evidence for a family/genus relationship than merely seeing a plant name in the index.

***Solanum nigrum*:** The official TKDL-linked 202041053693 examination material includes *Solanum nigrum* and *Withania somnifera* together in a claimed herbal composition for respiratory/COVID-related use and discusses traditional-use evidence for the constituents. This makes it a useful same-family candidate, but not automatically an equivalent substitute.

Official TKDL examination PDF: https://www.tkdil.res.in/tkdil/Outcomes/202041053693_Examiner.pdf

***Solanum surattense*:** Official TKDL examination material 1114/CHE/2010 identifies a single-ingredient formulation containing *Solanum surattense* for cough/bronchitis and also describes a compound formulation containing *Solanum surattense*, *Justicia adhatoda* and *Ocimum*

sanctum for respiratory-system diseases. This gives formulation-level evidence for the candidate.

Official TKDL record: https://www.tkdل.res.in/tkdل/LangGerman/common/Examinarreport/reports/1114_CHE_2010.asp

Solanum indicum: The official TKDL plant-name index verifies the entry as Bhat. For this first-pass corpus, we retain it as a same-family candidate but do not fabricate a formulation match without a specific public TKDL record.

6. Alternative-ranking example

User: “My Ashwagandha-based formulation has a prior-art concern.”

IP-SAKTI: identify Withania somnifera → identify Solanaceae → retrieve TKDL candidates → compare plant part, indication and formulation role → rank candidates.

First candidate to investigate: Withania coagulans, because the TKDL-linked examination evidence explicitly connects it to Withania somnifera. **Next candidates:** Withania ashwagandha, Solanum indicum, Solanum nigrum and Solanum surattense, with ranking adjusted as formulation/use evidence is retrieved.

7. Model-ready dataset fields

family | anchor_plant | candidate_plant | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent_IP_flag | evidence_level | verification_status

8. AI / legal guardrails

- Same family/genus does not mean plants are interchangeable.
- Same family/genus does not establish freedom from patent infringement.
- Never output “no legal barriers” without a separate claim-level patent/legal analysis.
- Distinguish traditional-knowledge evidence, novelty, inventive step and Section 3(p).
- Every recommendation should show the exact TKDL source and evidence level.
- Recommended wording: “**Potential alternative candidate for further research — botanical, formulation, efficacy and patent/legal verification required.**”

9. Verification note

TKDL states that its representative public database contains about 1,250 formulations selected from classical Ayurveda, Unani, Siddha and Sowa Rigpa texts. The public plant-name index and public examination/outcome records are therefore useful for building a verified demonstration corpus, but they are not equivalent to unrestricted access to the full TKDL database.

Official TKDL database information: <https://www.tkdل.res.in/tkdل/langdefault/common/More.asp?GL=Eng>